Pro Football Sports Science Research Update

March 2024

A mixed methods investigation into athletic trainer and dietitian's nutrition practices for sport-related concussion patients. Journal of Concussion, Vol. 7: 1-12, 2023.

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A review of the personality traits in post-concussion syndrome. Acta Neurologica Belgica, Ahead of Print: 1-12, 2024.

Acute effects of ad libitum use of commercially available cannabis products on the subjective experience of aerobic exercise: A crossover study. Sports Medicine, Ahead of Print: 1-16, 2023.

An analysis of positional generic and individualized speed thresholds within the most demanding phases of match play in the English Premier League. International Journal of Sports Physiology and Performance, Ahead of Print: 1-11, 2023.

Are all hamstring injuries equal? A retrospective analysis of time to return to full training following BAMIC type 'c' and T-junction injuries in professional men's rugby union. Scandinavian Journal of Medicine & Science in Sports, 34: e14586, 2024.

Association between physical demands, skin temperature and wellbeing status in elite football players. Scientific Reports, 13:13780, 2023.

Association of use of a mobile tackling dummy during college football practice with reduced sport-related concussion – Results of a pilot investigation. The Orthopaedic Journal of Sports Medicine, 12(2): 1-8, 2024.

Can player tracking devices monitor changes in internal response during multidirectional running? Research Quarterly for Exercise and Sport, 94(4): 975-981, 2023.

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Pro Football Sports Science Research Update

Cardiovascular disease in retired NFL players: A systematic review. The Physician and Sportsmedicine, Ahead of Print: 1-25, 2024.

Common body temperature sites provide invalid measures of body core temperature in hyperthermic humans wearing American football uniforms. Temperature, Vol, 8(2): 166-175, 2021.

Comparative analysis of patellar tendon, Achilles tendon and plantar fascia structure in indoor and outdoor football players: A novel cross-sectional pilot study. Scientific Reports, 14:3930, 2024.

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Pro Football Sports Science Research Update

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Pro Football Sports Science Research Update

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Pro Football Sports Science Research Update

The effects of repeated-sprint training on physical fitness and physiological adaptation in athletes: A systematic review and meta-analysis. Sports Medicine, Ahead of Print: 1-22, 2023.

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Pro Football Sports Science Research Update

A mixed methods investigation into athletic trainer and dietitian's nutrition practices for sport-related concussion patients. Journal of Concussion, Vol. 7: 1-12, 2023.

Background: Nutrients may have implications following concussion, such as inflammation reduction or neuroprotection. The purpose of this study was to describe nutrition practices of athletic trainers (AT) and dietitians for patients regarding sport-related concussion, with respect to prevention, recommendations, and barriers to implementation.

Hypotheses: Due to the descriptive nature of this study, hypotheses were not warranted.

Study Design: Mixed methods.

Level of Evidence: Level 3.

Methods: A survey was distributed through professional memberships and included multiple-choice and open-ended items about nutritional practices regarding concussions. Participants included 1465 ATs and 51 dietitians accessing the survey (AT: age =35.7±11.4 years, females =43.1%; dietitians: age =31.8±8.17 years, females =45.1%). Open-ended survey responses regarding nutritional practices for concussion prevention, recommendations, and changes following concussion were analyzed by identifying key words or repeating phrases, theme development and categories into an initial codebook, using the codebook on responses, and coding all responses with the finalized codebook. Barriers to implementation was analyzed with descriptive statistics.

Results: Themes were identified for prevention strategies, and recommendations and changes following concussion. For concussion prevention, ATs noted overall health, specific target nutrients, barriers and limited research support, while dietitians identified overall health, specific target nutrients, recommendations/guidelines as themes. Post-injury themes for patient recommendations and changes were noted as overall health, specific target nutrients, resources, interprofessional collaboration by ATs, whereas dietitians identified overall health, specific target nutrients, interprofessional collaboration. ATs and dietitians noted numerous barriers including the lack of clinician knowledge and athlete compliance.

Conclusions and Clinical Relevance: ATs and dietitians are utilizing nutritional practices regarding concussion in numerous ways by focusing on use for whole body, brain health and specific targets such

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as omega fatty acids, hydration, micronutrients/supplements, and macronutrients. With these findings and the barriers listed by respondents, we can use this information to design focused research to influence evidence-based nutritional practices and concussion.

A musculoskeletal multifactorial individualized program for hamstring muscle injury risk reduction in professional football: Results of a prospective cohort study. BMJ Open Sport & Exercise Medicine, 10: e001866, 2024.

Objective: To test whether a musculoskeletal multifactorial and individualized hamstring muscle injury (HMI) risk reduction program could reduce HMI risk in professional football.

Methods: We conducted a prospective cohort study in Finnish premier football league teams, with the 2019 season used as a control and an intervention conducted in the 2021 season. Screening was conducted to provide individualized programs and monitor progress. Cox regression with hazard ratio (HR) was used with HMI as outcome and season as explanatory variable, including all players for primary analysis and those who performed the two seasons for secondary analysis.

Results: 90 players were included in the control and 87 in the intervention seasons; 31 players performed in the 2 seasons. Twenty HMIs were recorded during the control and 16 during the intervention seasons. Cox regression analyses revealed that HMI risk at any given time was not significantly different between control and intervention seasons (for all players: HR 0.77 (95% CI 0.39 to 1.51), p=0.444; for the 31 players: HR 0.32 (95% CI 0.01 to 1.29), p=0.110)). For the 31 players, the HMI burden was significantly reduced in the intervention compared with the control season (RR 0.67 (95% CI 0.53 to 0.85)). Higher compliance with knee strength training, maximal velocity exposure and lower performance reductions in maximal theoretical horizontal force and knee flexor force were associated with lower HMI incidence.

Conclusions: Although the primary analysis did not reveal any significant effect of the intervention to reduce HMI risk in professional football, the program was feasible, and additional secondary analyses showed a significant association between the intervention and lower HMI burden, incidence and risk.

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WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Hamstring muscle injuries (HMIs) remain the most common injury in modern professional football.

⇒ Most successful HMI risk reduction interventions in professional football have mainly focused on only one component and on eccentric knee flexor strength.

⇒ It is relatively agreed upon that not reduced HMI rates could be partly due to the lack of multifactorial approaches and issues with compliance.

WHAT THIS STUDY ADDS

⇒ The innovative musculoskeletal multifactorial and individualized HMI risk program was feasible in a professional football setting.

⇒ The HMI risk at any given time was not significantly different between the control and the intervention seasons.

⇒ When considering only players who performed the two seasons, the HMI burden was significant ly reduced in the intervention season.

⇒ Higher compliance with knee strength training and maximal velocity exposure was associated with lower HMI incidence.

⇒ Decreases in maximal theoretical horizontal force and knee flexor force during the season increased the odds of sustaining an HMI.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ These promising results support the interest in disseminating the present musculoskeletal multifactorial and individualized program and promoting its use in football. Although it was conceived to be multifactorial, end-users could also consider using, more specifically, some components of the program which have been associated with lower HMI incidence and risk (ie, knee strength training, maximal theoretical horizontal force and maximal velocity exposure).

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Pro Football Sports Science Research Update

A review of the personality traits in post-concussion syndrome. Acta Neurologica Belgica, Ahead of Print: 1-12, 2024.

Introduction: Various manifestations ranging from physical symptoms to cognitive and emotional impairments could often be seen following head concussions that lead to mild traumatic brain injury (mTBI). These symptoms are commonly comprising the post-concussion syndrome (PCS) and their resolution could be influenced by multiple factors. Personality traits have been suggested as potential risk factors for the emergence and persistence of PCS. In this study, we aimed to investigate the possible predisposition to PCS given by certain personality traits.

Methods: Prospective cohort studies, observational studies, and cross-phenotype polygenic risk score analyses were selected from the main scientific databases (PubMed/Medline, Scopus, EMBASE, Web of Science) based on multiple-step screening, using keywords (such as "personality traits", "post-concussion syndrome", "traumatic brain injury", "anxiety", "depression", "resilience", and "somatization") and inclusion/exclusion criteria (English written studies available in full text presenting relevant data on TBI patients and their personality traits; reviews, animal studies, and studies not written in English, not available in full text, or not presenting full demographical and clinical data were excluded). The investigated personality traits included emotional reserve, somatic trait anxiety, embitterment, mistrust, parental anxiety, state anxiety, trait anxiety, anxiety sensitivity, pain catastrophizing, helplessness, sports-concussion symptom load, and cognitive resilience.

Results: The reviewed data from 16 selected studies suggested that personality traits play an essential role in the development and persistence of PCS. Emotional reserve, cognitive resilience, and lower levels of somatic trait anxiety were associated with better outcomes in PCS. However, higher levels of anxiety sensitivity, pain catastrophizing, helplessness, and sports-concussion symptom load were associated with worse outcomes in PCS. Parental anxiety was not associated with persistent symptoms in children following concussion. Despite the statistical analysis regarding the included publications bias was low, further studies should further investigate the correlation between TBI and some personality traits, as some of the selected studies did not include healthy individuals and their psychological profiles for comparison and correlation analysis.

Conclusion: Personality traits may help predict the development and persistence of PCS following mTBI. Understanding the personality traits roles in PCS could assist the development of targeted interventions

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for the prevention and treatment of PCS. Further research is needed to better understand the complex interactions between personality traits, neurobiological factors, and psychosocial factors in PCS.

Acute effects of ad libitum use of commercially available cannabis products on the subjective experience of aerobic exercise: A crossover study. Sports Medicine, Ahead of Print: 1-16, 2023.

Objective: The present study aimed to examine the acute effects of legal-market cannabis on regular cannabis users' subjective responses to exercise in a controlled laboratory environment.

Background: Given the stereotype that cannabis is associated with extreme sedentary behavior, there are concerns that cannabis legalization may exacerbate the US physical inactivity epidemic. However, despite these concerns, recent years have seen considerable public interest in the use of cannabis concurrently with exercise (e.g., running).

Methods: The present study compared participants' experiences of exercise without cannabis to their experiences of exercise after acute ad libitum use of one of two commercially available cannabis flower products: a Δ9-tetrahydrocannabinol-dominant or a cannabidiol-dominant product. Participants (N = 42) were regular cannabis users between the ages of 21 and 39 years (mean = 30.81 years, standard deviation = 4.72 years).

Results: Although participants reported a more positive affect ($p < 0.001$), enjoyment ($p < 0.001$), and runner's high symptoms ($p < 0.001$) during their cannabis (vs non-cannabis) exercise appointment, they also reported more exertion ($p = 0.04$). Pain levels were very low and did not differ between appointments ($p = 0.45$). Effects appeared to depend, in part, on cannabinoid content; there was a larger difference in enjoyment ($p = 0.02$), and a smaller difference in exertion ($p = 0.02$), between the cannabis and non-cannabis exercise appointments among participants in the cannabidiol (vs Δ9-tetrahydrocannabinol) condition.

Conclusions: To our knowledge, this is the first study to investigate the acute effects of commercially available cannabis on subjective responses to exercise in a laboratory environment. Our findings suggest that, among regular cannabis users who use cannabis in combination with exercise, cannabis use prior

to exercise may lead to increases in both positive and negative aspects of the subjective exercise experience. Research using diverse samples, exercise modalities, and methodologies (e.g., placebo-controlled trials) is needed to establish the generalizability of these findings.

Key Points

- In a sample of individuals who use cannabis with exercise, participants reported higher levels of positive mood states (e.g., enjoyment) when exercising under the influence of cannabis, regardless of whether they used a Δ9-tetrahydrocannabinol-dominant or cannabidiol dominant product.
- At the same time, participants who used a Δ9-tetrahydrocannabinol-dominant product indicated that exercise felt more effortful after cannabis use.
- These findings suggest that, among individuals who use cannabis with exercise, acute cannabis use may have both positive and negative impacts on the exercise experience.

An analysis of positional generic and individualized speed thresholds within the most demanding phases of match play in the English Premier League. International Journal of Sports Physiology and Performance, Ahead of Print: 1-11, 2023.

Objectives: To analyze the positional distances covered above generic and individualized speed thresholds within the most demanding phases of match play. Categorized by position, 17 English Premier League players' match data were analyzed over 2 consecutive seasons (2019–20 and 2020–21). The most demanding phases of play were determined using a rolling average across 4 periods of 1, 3, 5, and 10 minutes. Distance covered in the time above the standard speed of 5.5 m/s was analyzed, with individualized metrics based on the maximal aerobic speed (MAS) test data.

Results: Central defenders displayed lower values for high-intensity periods when compared with fullbacks, midfielders, and wide midfielders for both generic and individualized metrics. MAS during 1-minute periods was significantly higher for forwards when compared with central defenders (82.9 [18.9] vs 67.5 [14.8] for maximum high-speed running [HSR] and 96.0 [15.9] vs 75.7 [13.8] HSR for maximum

MAS activity). The maximum effect size differences between the central midfielders, wide midfielders, and fullbacks for HSR and MAS measures under the maximum HSR criterion was 0.28 and 0.18 for the 1-minute period, 0.36 and 0.19 for the 3-minute period, 0.46 and 0.31 for the 5-minute period, and 0.49 and 0.315 for the 10-minute period.

Conclusions: Individualized speed metrics may provide a more precise and comparable measure than generic values. Data appear to be consistent across playing positions except for central defenders. This information may allow practitioners to directly compare individualized physical outputs of non–central defenders during the most demanding phases of play regardless of the players' positional group. This may provide coaches with important information regarding session design, training load, and fatigue monitoring.

Are all hamstring injuries equal? A retrospective analysis of time to return to full training following BAMIC type 'c' and T-junction injuries in professional men's rugby union. Scandinavian Journal of Medicine & Science in Sports, 34: e14586, 2024.

Abstract - We aimed to determine whether the anatomical location (intramuscular tendon or T-Junction) of hamstring muscle injuries in professional men's rugby union associates with a prolonged time to return to full training and a higher rate of re-injury/subsequent injury. We reviewed the medical records of an Irish professional rugby union club to identify hamstring muscle injuries incurred across five seasons. Clinicians and players were not blinded to MRI results at the time of rehabilitation. A blinded musculoskeletal radiologist re-classified all included injuries (n=91) according to the British Athletics Muscle Injury Classification framework. Players who sustained an injury with intramuscular tendon involvement required a longer time to return to full training compared to players who sustained an injury without intramuscular tendon involvement (78days vs. 24days). Players who sustained a biceps femoris injury with T-junction involvement did not require a longer time to return to full training compared to players who sustained a biceps femoris injury without T-junction involvement (29days vs. 27days). Injuries with either intramuscular tendon or T-junction involvement were not associated with an increased rate of re-injury/subsequent injury to the same limb (intramuscular tendon involvement –

odds ratio=0.96, T-junction involvement−odds ratio=1.03). When a hamstring muscle injury involves the intramuscular tendon, the injured player and stakeholders should be made aware that a longer time to return to full training is likely required. T-junction involvement does not alter the expected clinical course of biceps femoris injuries.

Association between physical demands, skin temperature and wellbeing status in elite football players. Scientific Reports, 13:13780, 2023.

Abstract - The demanding nature of elite football requires players to be closely monitored to ensure optimal performance and minimize injury risk. This study aimed to evaluate the relationship between physical demands, skin temperature, and well-being status in 30 elite football players over a 12-week competitive period. Thermography assessments, weekly Well-being questionnaires, and daily training and match load recordings were used to gather data. Results indicated that along the microcycles there was a decrease in high-intensity accelerations and decelerations distance completed, while maintaining other high-intensity actions. Furthermore, it was found that high-intensity movements contribute to the generation of thermal asymmetries in the thighs; the adductor thermal asymmetry showed a positive relationship with stress and muscle soreness, the knee thermal asymmetry had a positive relationship with fatigue and a negative relationship with rest and quality of rest, and finally the hamstrings muscles exhibited significant differences between the thermal asymmetry groups, with the high asymmetry completing less high intensity actions than the low asymmetry group. In conclusion, this study highlights the interconnections between physical demands, skin temperature, and well-being in elite football players and provides valuable insights for coaches and trainers in their efforts to optimize performance and health.

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Pro Football Sports Science Research Update

Association of use of a mobile tackling dummy during college football practice with reduced sport-related concussion – Results of a pilot investigation. The Orthopaedic Journal of Sports Medicine, 12(2): 1-8, 2024.

Background: Considering the multifaceted consequences of improperly managed sport-related concussions (SRCs) in American football, identifying efficacious prevention measures for enhancing player safety is crucial.

Purpose: To investigate the association of primary prevention measures (no-tackle practices and using a mobile tackling dummy in practice) with the frequency of SRCs within college football programs in the United States.

Study design: Descriptive epidemiology study.

Methods: In this pilot study, we analyzed the frequency of new SRCs recorded during various settings (total, in preseason, in season, in practice, and game) across 14 seasons (2007-2019 and 2021) for Dartmouth College and across 7 seasons (2013-2019) for the 7 other teams in the Ivy League men's athletic football conference. Trends between seasons and the number of SRCs sustained were examined using correlations and basic descriptive statistics. We also examined SRC frequency in relation to primary prevention measures (no-tackle practices, use of mobile tackling dummies during practice) in the Dartmouth College football program, and we compared SRCs with regard to the no-tackle practice policy in the other Ivy League teams.

Results: There was a statistically significant reduction in the number of SRCs over the seasons studied, with the strongest finding observed for Dartmouth College in-game SRCs ($r = -0.52$; $P = .029$). Relatedly, the strongest between-season effect was seen for the Dartmouth College practice policy on in-game SRCs ($\eta 2 = 0.510$; $P = .01$). The use of mobile tackling dummies was found to be independently associated (adjusting for no-tackle practice) with a lower number total ($\beta = -0.53$; $P = .049$), in-season ($\beta = -0.63$; $P = .023$), and in-game ($\beta = -0.79$; $P = .003$) SRCs. While seasons with the no-tackle practice were not meaningfully associated with SRCs for Dartmouth College, stronger trends were observed in the other Ivy League teams, such that seasons with this policy were associated with lower SRC prevalence.

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Pro Football Sports Science Research Update

Conclusion: Our data indicate that the use of the mobile tackling dummy in practice was related to the reduced number of SRCs sustained at multiple settings during the football season. To a lesser extent, the no-tackle practice policy was also associated with a reduced number of SRCs.

Can player tracking devices monitor changes in internal response during multidirectional running? Research Quarterly for Exercise and Sport, 94(4): 975-981, 2023.

Purpose: We examined the movement, physiological and muscle function responses to running with and without (i.e. linear) multiple directional changes to understand which measures of external demands better reflected changes in the internal response.

Methods: Twelve team sport athletes completed a linear and multidirectional running trial during which movement characteristics, oxygen consumption (V_O2), blood lactate (B[La]) and heart rate (HR) were measured. Isometric peak torque of knee extensors and flexors was also assessed before and after each trial.

Results: High speed running distance was higher during the linear trial ($p < 0.001$), whereas time at high metabolic power ($p = 0.046$), number of accelerations ($p < 0.001$), summated HR ($p = 0.003$) and B[La] ($p = 0.002$) were higher during the multi-directional trial. Integrated external to internal ratios of high-speed running: summated HR and high-speed running: total V_O2 were different between multidirectional and linear trials ($p \leq 0.001$). Conversely, high metabolic power: summated HR and high metabolic power: total V_O2 were similar ($p \geq 0.246$). Small decrements in knee flexor ($p = 0.003$) and extensor torque ($p = 0.004$) were observed after both trials.

Conclusion: Time at high metabolic power better reflects the increased internal response during running with more directional changes than high speed running.

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Pro Football Sports Science Research Update

Cardiovascular disease in retired NFL players: A systematic review. The Physician and Sportsmedicine, Ahead of Print: 1-25, 2024.

Objective: Despite robust research endeavors exploring post-play health implications in former NFL players, the impact of former-player status on long-term cardiovascular health has not yet been elucidated. The purpose of this systematic review is to describe the available research on the cardiovascular health in former NFL players.

Methods: Relevant studies were included from the PubMed, Scopus, and Embase databases. Studies were evaluated in accordance with PRISMA guidelines. Two independent reviewers conducted the title/abstract screenings and risk of bias determinations. The results of the studies were extracted for inclusion in the review.

Results: Sixteen studies met inclusion criteria. Though evidence was discordant among studies, former NFL players appeared to possess more favorable metabolic profiles and decreased mortality compared to community controls. Of note, 90% of former players were found to be overweight or obese.

Conclusion: Though cardiovascular disease is the leading cause of death among former NFL players, they possess comparable metabolic and cardiovascular profiles to community controls. Further research is necessary to ascertain the impact of NFL play on cardiovascular health and develop tailored preventative care strategies for former players.

Common body temperature sites provide invalid measures of body core temperature in hyperthermic humans wearing American football uniforms. Temperature, Vol, 8(2): 166-175, 2021.

Abstract - Valid body core temperature measurements are essential for diagnosing and monitoring athletes with exertional heat stroke (EHS). Experts question the validity of body temperature sites that vary by >±0.27°C from the gold standard, rectal temperature (TREC). No research has established the

validity of body temperature sites when American football uniforms (PADS) are worn during simulated EHS scenarios. Thirteen men (age, 22 ± 2 y; mass, 77.5 ± 8.8 kg; height, 181.3 ± 5.7 cm) donned PADS

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and entered an environmental chamber (38.7 ± 0.8°C, 38.9 ± 2.4% relative humidity). We compared TREC to a forehead liquid crystal temperature monitor (TFHD), axillary (TAXL), oral (TORL), and aural temperatures (TEAR) 34 times over four consecutive periods: 10-minutes of rest; exercise until participants TREC was between 39.7°C and 39.8°C; cold-water immersion (CWI, 10.0 ± 0.1°C) until all temperature sites indicated ≤38°C; and a 15-minute post-immersion recovery period. Body temperatures varied between sites during all periods ($F_{36,432} \geq 2.5$, $P \leq 0.001$). TAXL and TORL statistically differed from TREC and exceeded the 0.27°C bias threshold at all 34 measurement times (100%). TFHD differed from TREC eight times during rest; eight times during exercise; 0 times during CWI; and twice during recovery (18/34, 53%). TFHD exceeded the bias threshold 28 times (82%). TEAR differed from TREC five times during rest; 0 times during exercise; five times during CWI; and once during recovery (11/34, 32%). TEAR exceeded the 0.27°C bias threshold 15 times during testing (44%). TAXL, TFHD, TEAR, and TORL should not be used to diagnose or monitor American football players with EHS.

Abbreviations ANOVA: analysis of variance; CWI: cold water immersion; EHS: exertional heat stroke; PADS: full American football uniform consisting of a helmet; shoulder, knee, thigh, hip and tailbone pads; a jersey top; undergarments; and half-length pants; TAXL: axillary temperature; TEAR: aural temperature; TFHD: liquid crystal temperature monitor; TORL: oral temperature; TREC: rectal temperature

Comparative analysis of patellar tendon, Achilles tendon and plantar fascia structure in indoor and outdoor football players: A novel cross-sectional pilot study. Scientific Reports, 14:3930, 2024.

Different sport modalities were associate with tendon adaptation or even tendon disturbances, such as volleyball, soccer or basketball.

Purpose: the aim of the present study was to determine the difference between indoor and outdoor football players on patellar tendon (PT), Achilles tendon (AT), plantar fascia (FP) and Hofa's fat pad thickness assessed with ultrasound imaging (USI). A cross-sectional study was developed with a total sample of 30 soccer players divided in two groups: outdoor group (n= 15) and indoor group (n= 15). The

thickness of PT, AT, PF and Hofa's fat pad has been assessed with USI. Hofa's fat pad reported significant differences for the left side between groups (P= 0.026). The rest of variables did not show any significant difference (P< 0.05). The ultrasonography assessment of the thickness of the PT, AT and PF did not show differences between outdoor and indoor football players. Hofa's fat pad resulted showed a significant decrease for outdoor soccer players with respect futsal players. Thus, it can be considered that the load stimuli received in both soccer players were not enough to produce structural adaptations in PT, AT and PF tissues.

Counteracting mental fatigue for athletes: A systematic review of the interventions. BMC Psychology, 12:67, 2024.

Abstract - The deleterious effects of mental fatigue (MF) on athletes have been carefully studied in various sports, such as soccer, badminton, and swimming. Even though many researchers have sought ways to ameliorate the negative impact of MF, there is still a lack of studies that review the interventions used to counteract MF among athletes. This review aims to report the current evidence exploring the effects of interventions on MF and sport-specific performance, including sport-specific motor performance and perceptual-cognitive skills. Web of Science, Scopus, PubMed, and SPORTDicus (EBSCOhost) were combed through to find relevant publications. Additionally, the references and Google Scholar were searched for any grey literature. For the current review, we included only randomized controlled trials that involved athletes, a primary task to induce MF, interventions to counter MF with comparable protocols, and the outcomes of sport-specific motor performance and perceptual-cognitive skill. The selection criteria resulted in the inclusion of 10 articles. The manipulations of autonomous self-control exertion, person-ft, nature exposure, mindfulness, and transactional direct current stimulation showed that positive interventions counteract MF and improve sport-specific performance in different domains, including strength, speed, skill, stamina, and perceptual-cognitive skills. The selected interventions could significantly counteract MF and improve subsequent sport-specific performance. Moreover, self-regulation and attention resources showed the importance of the potential mechanisms behind the relevant interventions.

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Current Concepts in Achilles Tendon Ruptures. Clinics in Podiatric Medicine and Surgery, 41: 153-168, 2024.

KEY POINTS

- Acute Achilles tendon ruptures can be treated either conservatively or surgically, both with functional rehabilitation and early range of motion.
- Conservative therapy has evolved from immobilization with strict non-weight-bearing protocol, to early functional rehabilitation with earlier weight-bearing.
- Surgical treatment approaches vary from percutaneous, minimally invasive, and open repair.
- Current literature discussing surgical treatment for ruptured tendons shows low incidence of complications and similar functional outcomes

Does wet-cupping therapy improve repeated sprint ability, perceived wellness, and rating of perceived exertion in young active males? Sports Medicine and Health Science, Ahead of Print: 1-7, 2023.

Abstract - Wet-cupping therapy (WCT) is one of the oldest known medical techniques, used as a traditional and complementary therapy with a wide application all around the world for general health. Research on the effects of WCT on sports performance are sparse and inconsistent. Thus, we aimed to explore the effects of WCT on repeated sprint ability, wellness, and exertion in young active males. Forty-nine active adult males (age: [28 ± 5] years; body height [177 ± 8] cm; body mass: [79 ± 7] kg; body mass index: [25.4 ± 1.8] kg/m^2) were selected for the study. The participants performed a running-based sprint test on two separate occasions (Control and Post-WCT). WCT was performed 24 h before the testing session. They completed the Hooper questionnaire to assess their well-being (i.e., sleep, stress, fatigue, and soreness) before each session. The rating of perceived exertion (RPE) was collected after each testing session. A higher maximum power ($p < 0.05$, effect size [ES] = 0.6), mean power ($p < 0.01$, ES = 0.5) and minimum power ($p < 0.01$, ES = 0.6) were recorded post-WCT as compared to Control session along with a better perceived sleep ($p < 0.01$, ES ¼ 0.85). Perceived stress ($p < 0.01$, ES = 0.6) and RPE ($p < 0.001$; ES = 1.1) were lower during the post-WCT compared to the Control session. The present

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findings demonstrated that WCT moderately enhanced repeated sprint ability and had positive effects on perceived sleep, stress, and exertion. WCT may be an effective ergogenic aid to improve repeated sprint ability and general well- being in young adult males. Future large-scale multicentric clinical studies are paramount to confirm the results of our study.

Fiber-type traps: Revisiting common misconceptions about skeletal muscle fiber types with application to motor control, biomechanics, physiology, and biology. Journal of Applied Physiology, 136: 109-121, 2024.

Abstract - Skeletal muscle is a highly complex tissue that is studied by scientists from a wide spectrum of disciplines, including motor control, biomechanics, exercise science, physiology, cell biology, genetics, regenerative medicine, orthopedics, and engineering. Although this diversity in perspectives has led to many important discoveries, historically, there has been limited overlap in discussions across fields. This has led to misconceptions and oversimplifications about muscle biology that can create confusion and potentially slow scientific progress across fields. The purpose of this synthesis paper is to bring together research perspectives across multiple muscle fields to identify common assumptions related to muscle fiber type that are points of concern to clarify. These assumptions include 1) classification by myosin isoform and fiber oxidative capacity is equivalent, 2) fiber cross-sectional area (CSA) is a surrogate marker for myosin isoform or oxidative capacity, and 3) muscle force-generating capacity can be inferred from myosin isoform. We address these three fiber-type traps and provide some context for how these misunderstandings can and do impact experimental design, computational modeling, and interpretations of findings, from the perspective of a range of fields. We stress the dangers of generalizing findings about “muscle fiber types” among muscles or across species or sex, and we note the importance for precise use of common terminology across the muscle fields.

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Frontal plane pelvic kinematics during high velocity running: Association with hamstring injury history. Physical Therapy in Sport, 64:133-139, 2023.

Introduction: Hamstring injuries are the most prevalent non-contact soft tissue injury in sports, with a larger portion of injuries being recurrent. The sagittal plane running kinematics correlated to hamstring injury history has been well documented. However, analysis of frontal plane kinematics allows for observation of stability and symmetry. This study aimed to examine the frontal plane running kinematics of elite collegiate level sprinters, with and without previous hamstring injury, compared to healthy counterparts.

Methods: Thirty-nine participants performed three 50-m sprints, with three inertial measurement unit sensors placed on the pelvis: one on each iliac crest and one on the sacrum. Participants were classified based on sex, competitive status, and injury history. To investigate differences based on group classification, the data were used to analyze mediolateral motion (relative magnitude of mediolateral acceleration) and asymmetry (difference in acceleration between right and left iliac crests) during each stance phase throughout the run.

Results: Injured sprinters displayed significantly greater mediolateral motion and asymmetry during stances than healthy counterparts.

Conclusions: This study demonstrates that frontal plane running stance dynamics are different in athletes with previous hamstring injury than healthy athletes. These athletes may benefit from rehabilitation strategies targeting postural control and stability during dynamic tasks.

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Fundamentals of individual loading for strength training after knee injuries. Sports Orthopaedics and Traumatology, 39:368-377, 2023.

Summary - The article highlights newer aspects of strength training after knee trauma that need to be considered in the future. Recent findings on muscle contraction (three-filament theory) and the importance of eccentric training for prevention and rehabilitation are discussed. The altered central, spinal, and peripheral neuromuscular activation patterns will be addressed after knee trauma and during eccentric muscle actions and their causes. Biomechanical references and psychological (anticipatory) aspects are discussed after knee trauma. In distinction to strength training with healthy persons, concrete hints for strength training and its load normative with persons after knee traumas are given. Finally, newer developments, including improved sensor technology, are presented, and concrete advice is given on optimizing strength training after knee joint injuries and surgery.

General manager and head coach exits in the NBA and the NFL. Sport, Business and Management, Vol. 14(1): 1-18, 2024.

Purpose: This article seeks to enhance the understanding as to why head coaches and general managers (GMs) in the National Basketball Association (NBA) and the National Football League (NFL) exit from their positions.

Design/methodology/approach: Three hypotheses were investigated using a series of quantitative and qualitative data from the past 30 years. The samples analyzed are comprised of 891 GM and coach annual observations for the NBA clubs and 949 GM and coach observations for the NFL clubs. Analyses include a logit analysis for coach exit/retention, a logit analysis for GM exit/retention and textual analysis via topic modeling via latent Dirichlet allocation.

Findings: Results show a correlation between a coach exiting and a GM exiting simultaneously, thus amplifying the importance of these two roles in enhancing or destroying the success of a club and supporting the need for a deeper understanding of both roles, particularly the GM. The results further highlight cultural differences across clubs in terms of GM and coach turnover, a factor that often is heavily influenced by club ownership.

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Originality/value: The results support the role of owners in exits, confirm the importance of winning in avoiding an exit, find a high level of interrelationship between GM and coach exits and show that past culture of firings influences future exit decisions.

Genetic variants in myostatin and its receptors promote elite athlete status. BMC Genomics, 24:761, 2023.

Background: While product of the myostatin gene (MSTN) is an important factor influencing muscle growth, which is well confirmed in nonhuman species, it has not been clearly confirmed whether MSTN expression influences interindividual differences in skeletal muscle mass, affects posttraining changes, or plays a role in the age-related loss of muscle mass and function in humans. Although the inconclusive results are usually explained by ethnic differences and the low frequency of some alleles, it is possible that the role of receptors (ACVR2A and ACVR2B) that affect the biological activity of myostatin is crucial. Therefore, we investigated the sequences of the MSTN, ACVR2A, and ACVR2B genes and determined the interaction between allelic variants and athletic performance and competition level in the Caucasian population. One hundred-two athletes were recruited for the sequencing study, and whole-genome sequencing (WGS) was performed. Second, 330 athletes and 365 controls were included, and real-time PCR was performed.

Results: The sequence analysis revealed two polymorphisms relatively common in the athlete cohort, and the alternate allele showed overrepresentation in athletes: MSTN rs11333758 and ACVR2A rs3764955. Regarding the polymorphic site MSTN rs11333758, there was a significant overrepresentation of the -/- genotype in all high-elite and mixed-sport high-elite athletes. Carriers of the ACVR2A rs3764955 CC and GG genotypes were more likely to be elite and high-elite athletes. In addition, carriers of the CC genotype were more likely to be in the mixed-sport sub-elite group. The gene–gene interaction analysis revealed that mixed-sport high elite athletes showed significant underrepresentation of the ACVR2A rs3764955 GC - MSTN rs11333758 AA genotype combination. In the same group, we observed a significant overrepresentation of the ACVR2A rs3764955 GC - MSTN rs11333758 -/- and the ACVR2A rs3764955 CC - MSTN rs11333758 -/- genotype combinations.

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Conclusions: We showed that the specific genotypes of the MSTN rs11333758 and ACVR2A rs3764955, either individually or in gene–gene combination, are significantly associated with athletes' competition level in the Polish population, especially in the mixed-sports athlete group. Thus, although further research is required, these polymorphisms, alone or in combination with other polymorphisms, are among the numerous candidates that could explain individual variations in muscle phenotypes.

Gut check: Unveiling the influence of acute exercise on the gut microbiota. Experimental Physiology, Ahead of Print: 1-15, 2023.

Abstract - The human gastrointestinal microbiota and its unique metabolites regulate a diverse array of physiological processes with substantial implications for human health and performance. Chronic exercise training positively modulates the gut microbiota and its metabolic output. The benefits of chronic exercise for the gut microbiota may be influenced by acute changes in microbial community structure and function that follow a single exercise bout (i.e., acute exercise). Thus, an improved understanding of changes in the gut microbiota that occur with acute exercise could aid in the development of evidence-based exercise training strategies to target the gut microbiota more effectively. In this review, we provide a comprehensive summary of the existing literature on the acute and very short-term (<3 weeks) exercise responses of the gut microbiota and faecal metabolites in humans. We conclude by highlighting gaps in the literature and providing recommendations for future research in this area.

New Findings

· What is the topic of this review?

The chronic benefits of exercise for the gut microbiota are likely influenced by acute changes in microbial community structure and function that follow a single exercise bout. This review provides a summary of the existing literature on acute exercise responses of the gut microbiota and its metabolic output in humans.

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· What advances does it highlight?

Acute aerobic exercise appears to have limited effects on diversity of the gut microbiota, variable effects on specific microbial taxa, and numerous effects on the metabolic activity of gut microbes with possible implications for host health and performance.

High-speed running distance and frequency in football training: When and how are they stimulated in a microcycle? International Journal of Sports Science & Coaching, Vol. 18(4):1132-1141, 2023.

Purpose: The aims of this study were to describe the distribution of the distance and frequency of the high-speed running performed by professional players during the training process, taking into account the duration of the microcycle (Mi) composed of 4, 5, 6 and 7 days between matches (MD). Total distance at high speed (HSR, >21 km·h−1, in m), at very high-speed running (VHSR, >24 km·h^{-1}, in m) and sprint (SPR, >27 km·h^{-1}, in m) were recorded, in terms of distance (d) and frequency (n) during training sessions. Training data were obtained using global positioning technology from 23 professional football players belonging to the first team of the Spanish La Liga club during the 2017/18 season (n = 28 microcycles). There were no significant differences in volume (distance and frequency) of the different locomotive demands (LD) when Mi accumulated more days. Regardless of the duration of the Mi, the central days (MD-5, MD-4 and MD-3) accumulated more stimulus in HSR, VHSR and SPR compared to the post sessions (MD-6 and MD-7) and pre-match (MD-2 and MD-1) where the MD-2 was the day that required the least stimuli from the players ($p < 0.05$). The Mi7 presented significantly higher external load values in the MD-5 and MD-4 compared to the Mi5 and Mi4, where the MD-3 was the day that received the greatest stimulation in the locomotive demand ($p < 0.01$). The data demonstrate that the LD (HSR, VHSR and SPR), accumulated greater stimulus in the central days of Mi, far from the previous competition, favoring an optimal recovery without compromising performance for the next game.

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In professional football the decline in high-intensity running activities from first to second half is more pronounced in players with a fast muscle typology. Scandinavian Journal of Medicine & Science in Sports, Ahead of Print: 1-12, 2023.

Abstract - Muscle typology is heterogeneous among national level football (soccer) players, but positional differences remain unclear. Furthermore, fast typology (FT) individuals fatigue more than slow typology (ST) individuals in lab conditions. Therefore, we investigated if muscle typology is different between playing positions and if the decay in high-intensity activities from the first to the second half is larger in FT football players than in ST players. We estimated muscle typology in 147 male professional football players by measuring soleus and gastrocnemius muscle carnosine via proton magnetic resonance spectroscopy. Players were classified as ST, intermediate typology (IT) or FT and categorized as goalkeeper, center back, full back, midfielder, winger or forward. Across four seasons in-game distances covered in multiple running speed, acceleration and deceleration zones were collected during the first and second half. We found no differences in muscle typology between positions ($p = 0.412$). FT players covered 10.9% more high acceleration distance (>3 $m.s^{-2}$) in the first half than ST players ($p = 0.021$) and high acceleration distance decay was larger for FT players (-12.4%) than ST (-7.7%; $p = 0.006$) and IT players (-7.3%; $p = 0.010$). Moreover, the decline in distance covered in several high-intensity zones tended to be larger in FT players (-11.2% high-intensity >15 $km.h^{-1}$; -12.7% high deceleration <-3 $m.s^{-2}$; -11.5% medium acceleration 2-3 $m.s^{-2}$) than in ST players (-7.1% high-intensity; -8.1% high deceleration; -8.1% medium acceleration; $0.05 < p < 0.1$). In conclusion, possessing a particular muscle typology is not required to play any football position at the national level. However, there are indications that FT players might fatigue more toward the end of the game compared to ST players.

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Individual in-situ GPS-derived acceleration-speed profiling: Toward automatization and refinement in male professional rugby union players. Sports Medicine – Open, 10:6, 2024.

Background: Recently a proof-of-concept was proposed to derive the soccer players' individual in-situ acceleration-speed (AS) profile from global positioning system (GPS) data collected over several sessions and games. The present study aimed to propose an automatized method of individual GPS-derived in-situ AS profiling in a professional rugby union setting.

Method: AS profiles of forty-nine male professional rugby union players representing 61.5 million positions, from which acceleration was derived from speed during 51 training sessions and 11 official games, were analyzed. A density-based clustering algorithm was applied to identify outlier points. Multiple AS linear relationships were modeled for each player and session, generating numerous theoretical maximal acceleration (A0), theoretical maximal running speed (S0) and AS slope (ASslope, i.e., overall orientation of the AS profile). Each average provides information on the most relevant value while the standard deviation denotes the method accuracy. In order to assess the reliability of the AS profile within the data collection period, data were compared over two 2-week phases by the inter-class correlation coefficient. A0 and S0 between positions and type of sessions (trainings and games) were compared using ANOVA and post hoc tests when the significant threshold had been reached.

Results: All AS individual profiles show linear trends with high coefficient of determination ($r2 > 0.81$). Good reliability (Inter-class Correlation Coefficient ranging from 0.92 to 0.72) was observed between AS profiles, when determined 2 weeks apart for each player. AS profiles depend on players' positions, types of training and games. Training and games data highlight that highest A0 are obtained during games, while greatest S0 are attained during speed sessions.

Conclusions: This study provides individual in-situ GPS-derived AS profiles with automatization capability. The method calculates an error of measurement for A0 and S0, of paramount importance in order to improve their daily use. The AS profile differences between training, games and playing positions open several perspectives for performance testing, training monitoring, injury prevention and return-to-sport sequences in professional rugby union, with possible transferability to other sprint-based sports.

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Intersection of race and socioeconomic status on concussion recovery among NCAA student-athletes: A CARE Consortium Study. Medicine and Science in Sports and Exercise, Vol. 55(12): 2180-2193, 2023.

Purpose: The objectives of this study are to 1) describe collegiate student-athlete (SA) race and household income and 2) evaluate time to normal academic performance (i.e., return to learn (RTL)), initiation of the return to play (iRTP) protocol, RTP protocol duration, and time to unrestricted RTP (URTP) after sustaining sport-related concussion (SRC).

Methods: Data were collected between 2014 and 2020 by the Concussion Assessment, Research, and Education Consortium. Baseline data were used to characterize participant demographics (N = 22,819) and post-SRC outcomes (n = 5485 SRC) in time to RTL (n = 1724) and RTP outcomes (n = 2646) by race. Descriptive statistics and nonparametric tests examined differences across race by demographic and injury characteristics. Kaplan-Meier curves estimated median days to RTL, iRTP protocol, RTP protocol completion, and URTP by race and covariate measures. Multivariable Cox proportional hazards regression assessed the effect of race on risk of RTL and RTP recovery time points.

Results: SA largely identified as White (75%) followed by Black (14%), multiracial (7%), and Asian (3%). More than half (53%) of all SA reported a household income of >$120,000, whereas 41% of Black SA reported a household income <$60,000. Race was not associated with relative risk of RTL or iRTP but was associated with RTP protocol completion and URTP. Non-Black/non-White SA were 17% less likely (adjusted hazard ratio = 0.83; 95% confidence interval = 0.71, 0.97) to complete the RTP protocol, and Black SA were 17% more likely (adjusted hazard ratio = 1.17; 95% confidence interval = 1.05, 1.31) to reach the URTP time point compared with White SA.

Conclusions: The present findings suggest collegiate SA enrolled in the Concussion Assessment, Research, and Education Consortium are primarily White and come from household incomes well above the US median. Race was not associated with RTL or iRTP but was associated with RTP protocol duration and total time to URTP. Clinicians should be conscientious of how their implicit or preconceived biases may influence SRC management among National Collegiate Athletic Association SA.

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Is melatonin as an ergogenic hormone a myth? A systematic review and meta-analysis. Endocrine, Ahead of Print: 1-20, 2023.

Purpose: Melatonin supplementation has been disclosed as an ergogenic substance. However, the effectiveness of melatonin supplementation in healthy subjects has not been systematically investigated. The present study analyzed the effects of melatonin supplementation on physical performance and recovery. In addition, it was investigated whether exercise bout or training alter melatonin secretion in athletes and exercise practitioners.

Methods: This systematic review and meta-analysis were conducted and reported according to the guidelines outlined in the PRISMA statement. Based on the search and inclusion criteria, 21 studies were included in the systematic review, and 19 were included in the meta-analysis.

Results: Melatonin supplementation did not affect aerobic performance relative to time trial (-0.04; 95% CI: -0.51 to 0.44) and relative to VO2 (0.00; 95% CI: -0.57 to 0.57). Also, melatonin supplementation did not affect strength performance (0.19; 95% CI: -0.28 to 0.65). Only Glutathione Peroxidase (GPx) secretion increased after melatonin supplementation (1.40; 95% CI: 0.29 to 2.51). Post-exercise melatonin secretion was not changed immediately after an exercise session (0.56; 95% CI: -0.29 to 1.41) and 60 min after exercise (0.56; 95% CI: -0.29 to 1.41).

Conclusion: The data indicate that melatonin is not an ergogenic hormone. In contrast, melatonin supplementation improves post-exercise recovery, even without altering its secretion.

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Mental fatigue impairs repeated sprint and jump performance in team sport athletes. Journal of Science and Medicine in Sport, Ahead of Print: 1-21, 2023.

Objectives: We tested whether mental fatigue (MF), induced by a cognitively-demanding task, would impair repeated sprint ability (RSA) and repeated jump ability (RJA) performance, and whether physical fatigue and MF would impair psychomotor vigilance.

Design: Randomized within-participant design.

Methods: After establishing baseline peak countermovement jump (CMJ), 18 male participants performed 12 maximal 20-m (10-m linear + 10-m directional) repeated sprints (RSA random test) followed by 12 maximal repeated CMJs (RJA test) subsequent to 30-min Stroop task (MF) or a documentary (Control). Peak and mean running time and height, percent decrement score (Sdec), blood lactate, heart rate and RPE were measured for CMJ, RSA, and RJA tests. MF (M-VAS) and psychomotor vigilance [psychomotor vigilance test (PVT)] were measured at baseline, after each condition, and after the RSA/RJA tests.

Results: Compared to Control, the Stroop task elevated MF ($p = .001$), RPE ratings (all $p < .031$), and mean and Sdec performance in directional (but not linear) RSA (all $p < .032$) and RJA tests (all $p < .034$). PVT score worsened after Stroop task ($p = .011$) but not Control, declined after RSA/RJA tests in both conditions (all $p < .023$) and was lower in the MF condition ($p = .029$). No condition differences were noted for peak (CMJ, RSA and RJA tests) performance, blood lactate, and heart rate.

Conclusions: MF impairs directional RSA, and RJA performance. This impairment was linked with increased RPE and without physiological changes. The progressive impairment in PVT score suggests a cumulatively negative effect of mental and physical fatigue on psychomotor vigilance.

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Parametric modeling and analysis of NFL run plays. Journal of Sports Analytics, Ahead of Print: 1-16, 2023.

Abstract - The paper is concerned with the modeling of run plays from data obtained from the NFL. Using a parametric regression model based on the skew–t distribution we estimate the shifts from overall league averages for each team within the NFL. From the interpretation of the parameters we can investigate what the best teams are specifically doing to achieve better performance according to the criterion of average yards per play.

Performing Nordic hamstring strength testing with additional weight affects the maximal eccentric force measured: Do not compare apples to oranges. BMJ Open Sport & Exercise Medicine, 9:e001699, 2023.

Background: Nordic hamstring test devices are commonly used to measure maximal eccentric hamstring force. The ability to control the final phase of the exercise has been adopted as a criterion to add weight when testing, without substantial evidence. We investigated if adding weight affected the maximal force measured, and if there were differences between players who could and could not control the final phase.

Methods: Female (n=84) and male (n=56) football players performed a Nordic hamstring strength test with 0, 5 and 10 kg. We used visual inspection to assess the ability to control the final phase (approximately last 20°), as per previously published studies.

Results: Maximal force was higher when tested with 5 kg (females: +8 N (2%), $p<0.001$; males: +18 N (4%), $p<0.001$) and 10 kg (females: +17 N (5%), $p<0.001$; males: +27 N (6%), $p<0.001$) compared with 0 kg. This was the case for both groups, those who could control the final phase (5 kg: +16 N (4%), $p<0.001$; 10 kg: +28 N (7%), $p<0.001$) and those who could not (5 kg: +9 N (3%), $p<0.001$; 10 kg: +15 N (4%), $p<0.001$).

Conclusion: Both players who could and could not control the final phase of the Nordic hamstring test demonstrated higher maximal force when adding weight to testing. Therefore, this should not be used

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to decide if players should perform testing with or without weight. Either all participants or none should be tested with weight, and the same approach should be used both for pre-testing and post-testing.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Nordic hamstring strength testing is a common and reliable method to measure maximal eccentric ham string strength. However, several studies have added weight to Nordic hamstring testing if participants are able to control the forward falling movement during the final phase of the test, without substantial evidence.

WHAT THIS STUDY ADDS

⇒ This study showed that participants, regardless of their ability to control the final phase of the test, displayed higher maximal eccentric hamstring force when performing Nordic hamstring strength testing with weight compared with testing without weight.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ To avoid potentially overestimating the effect of a training intervention or causing biased result, future research should not use the ability to control the final phase of the Nordic hamstring test to decide if participants should be tested with added weight or not.

Personalized tendon loading reduces muscle-tendon imbalances in male adolescent elite athletes. Scandinavian Journal of Medicine & Science in Sports, 34:e14555, 2024.

Abstract - An imbalanced adaptation of muscle strength and tendon stiffness in response to training may increase tendon strain (i.e., the mechanical demand on the tendon) and consequently tendon injury risk. This study investigated if personalized tendon loading inducing tendon strain within the effective range for adaptation (4.5%-6.5%) can reduce musculotendinous imbalances in male adolescent handball athletes (15-16 years). At four measurement time points during a competitive season, we assessed knee extensor muscle strength and patellar tendon mechanical properties using dynamometry and

ultrasonography and estimated the tendon's structural integrity with a peak spatial frequency (PSF) analysis of proximal tendon ultrasound scans. A control group (n = 13) followed their usual training routine, an intervention group (n = 13) integrated tendon exercises into their training (3x/week for ~31 weeks) with a personalized intensity corresponding to an average of ~6.2% tendon strain. We found a significant time by group interaction ($p < 0.005$) for knee extensor muscle strength and normalized patellar tendon stiffness with significant increases over time only in the intervention group ($p < 0.001$). There were no group differences or time-dependent changes in patellar tendon strain during maximum voluntary contractions or PSF. At the individual level, the intervention group demonstrated lower fluctuations of maximum patellar tendon strain during the season ($p = 0.005$) and a descriptively lower frequency of athletes with high-level tendon strain (≥9%). The findings suggest that the personalized tendon loading program reduced muscle-tendon imbalances in male adolescent athletes, which may provide new opportunities for tendon injury prevention.

Prevalence of carboxy-Δ8 -tetrahydrocannabiniol in antidoping samples. Drug Testing and Analysis, Ahead of Print: 1-5, 2024.

Abstract - Δ9 -Tetrahydrocannabinol (Δ9 -THC) is usually the primary psychoactive agent in cannabis preparations. Recently, products containing another isomer, Δ8 -tetrahydrocannabinol (Δ8 -THC), have become available for sale. Δ8 -THC exists naturally in the cannabis plant at very low concentrations; hence, the Δ8 -THC present in most of the above-mentioned products is likely to be manufactured synthetically. A surge in popularity of these products, coupled with little oversight to ensure purity and potency, has led to reports of adverse events. Workplace drug testing programs as well as many sporting organizations prohibit the use of cannabinoids. Carboxy-Δ9 -THC (Δ9 -THC-COOH) is the targeted urinary metabolite for detection of cannabis use. The proliferation of products containing Δ8 -THC, which metabolizes to Δ8 -THC-COOH, presents analytical complexity with respect to separation and quantification of the individual isomers as well as legal complexity with respect to lack of clarity around the legal status of Δ8 -THC. This study aims to estimate the prevalence of Δ8 -THC use in the athlete community by monitoring for Δ8 -THC-COOH in samples collected for antidoping. A high-performance liquid chromatography tandem mass spectrometry (LC-MS/MS) method was utilized to resolve Δ8 and Δ9 -THC-COOH. One thousand samples with a presumptive Δ9 -THC-COOH finding in routine screening

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were analyzed by the above LC-MS/MS method. Approximately 12% of samples contained Δ8 -THC-COOH at relative abundances between 5% and 100% of total carboxy-THC content.

Public chronic traumatic encephalopathy knowledge: Sources, accuracy and confidence. Current Sports Medicine Reports, Vol. 23(1): 23-28, 2024.

Abstract - The goal of this study was to examine the general public's level of accuracy and confidence in knowledge of chronic traumatic encephalopathy (CTE), as well as information sources. This study also explored how these factors affected comfort in allowing children to play a high-contact sport. This study utilized online surveys and included 529 participants. Overall, CTE knowledge accuracy was 48.02% (standard deviation = 0.23). Inaccuracies regarding the etiology and diagnosis of CTE were most common, whereas the symptoms and lack of treatments for CTE were more widely known. Despite overall low CTE knowledge accuracy, CTE knowledge confidence was positively correlated with comfort in allowing children to play a high-contact sport ($r = 0.199$, $P \leq 0.001$). Participants identified television/movies followed by web sites and social media as the most utilized CTE information sources. These results further support the need for clinicians and researchers to address misconceptions about CTE.

Real-world fatigue testing in professional rugby union: A systematic review and meta-analysis. Sports Medicine, Ahead of Print: 1-20, 2023.

Background: Professional rugby union is a high-intensity contact sport with position-specific high training and match volumes across a season that may lead to periods of fatigue if above a typically experienced threshold. This study assesses the influence of match play and/or training on fatigue levels in rugby union players.

Objective: We aimed to perform a systematic review and meta-analysis of measures used to assess fatigue status in male professional rugby union players.

Pro Football Sports Science Research Update

Methods: Using electronic databases (PubMed, SPORTDiscus, Web of Science, Cochrane Library, EMBASE, and MEDLINE), a systematic review of fatigue testing in rugby union was conducted on (1) neuromuscular, (2) subjective self-report, (3) biochemical, and (4) heart rate-derived measures.

Results: Thirty-seven articles were included in this systematic review, of which 14 were further included in a meta-analysis. The results of the meta-analysis revealed small, yet not significant, decreases in countermovement jump height immediately after (effect size [ES] = - 0.29; 95% confidence interval [CI] - 0.64 to 0.06), 24 h (ES = - 0.43; 95% CI - 3.99 to 3.21), and 48 h (ES = - 0.22; 95% CI - 0.47 to 0.02) after exposure to rugby union match play or training. Reported wellness (ES = - 0.33; 95% CI - 1.70 to 1.04) and tiredness (ES = - 0.14; 95% CI - 1.30 to 1.03) declined over a period of a few weeks (however, the results were not-statistically significant), meanwhile muscle soreness increased (ES = 0.91; 95% CI 0.06 to 1.75) within the 96 h after the exposure to rugby union match play or training. Finally, while cortisol levels (ES = 1.87; 95% CI - 1.54 to 5.29) increased, testosterone declined (ES = - 1.54; 95% CI - 7.16 to 4.08) within the 24 h after the exposure. However, these results were not statistically significant.

Conclusions: Subjective measures of muscle soreness can be used to assess fatigue after match play and training in rugby union players. Within-study and between-study variability for countermovement jump height, biochemical markers, and heart rate-derived measures means the utility (practical application) of these measures to assess fatigue in professional rugby union players after matches and training is unclear.

Key Points

- Results indicate acceptable utility for measures of subjective soreness when assessing the influence of match play and/or training in professional rugby union.
- High inter-individual variability for assessing the influence of match play and/or training in professional rugby union was evident in countermovement jump, biochemical markers and heart rate-derived measures.
- Practitioners are advised to assess individual training response to match play and/or training in professional rugby union to better guide training prescription and optimize player readiness.

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Seasonal changes in free 25-(OH)D and vitamin D metabolite ratios and their relationship with psychophysical stress markers in male professional football players. Frontiers in Physiology, 14:1258678, 2023.

Introduction: Novel markers of vitamin D status are currently being investigated, including free 25-(OH)D (25-(OH)DF) and the vitamin D metabolite ratio (24,25- (OH)2D3:25-(OH)D3; VMR). The VMR may provide additional functional information on vitamin D metabolism in athletes. Therefore, the main objective of the current study was to evaluate 25-(OH)D_F, bioavailable 25-(OH)D (25-(OH)D_B), VMR, and psychophysical stress markers during different training periods over a half-season. The second aim was to assess the association between vitamin D binding protein (VDBP), total and free 25-(OH)D, VMRs, and psychophysical stress markers in professional football players. Moreover, we examined the relationship between 25-(OH)D_3 and vitamin D metabolites (24,25-(OH)2D_3, 3-epi-25-(OH)D_3) to determine if training loads in different training periods influenced the vitamin D metabolome.

Methods: Twenty professional football players were tested at six different time points across half a year (V1—June; V2—July; V3—August; V4—October; V5—December; V6—January).

Results: Analyses indicated a significant seasonal rhythm for VDBP, and total 25- (OH)D (25-(OH)DT), 25-(OH)DB, 24,25-(OH)2D_3, 3-epi-25-(OH)D_3, 25-(OH)D_3: 24,25-(OH)2D_3, and 24,25-(OH)2D_3:25-(OH)D_3 VMRs throughout the training period. No correlation was detected between 25-(OH)D_T, 25-(OH)D_B, 25-(OH) D_F, vitamin D metabolites, VMRs, VDBP, and ferritin, liver enzymes (aspartate transaminase [AST] and alanine transaminase [ALT]), creatine kinase (CK), cortisol, testosterone, and testosterone-to-cortisol ratio (T/C) in each period (V1-V6). However, there was a strong statistically significant correlation between 25- (OH)D_3 and 24,25-(OH)D_3 in each training period.

Conclusion: In conclusion, a seasonal rhythm was present for VDBP, 25-(OH)D_T, 25-(OH)DB, vitamin D metabolites (24,25-(OH)2D_3, 3-epi-25-(OH)D_3), and VMRs (25-(OH)D_3:24,25-(OH)2D_3, 25-(OH)D_3:3-epi-25-(OH)D_3). However, no rhythm was detected for 25-(OH)DF and markers of psychophysical stress (ferritin, liver enzymes, CK, testosterone, cortisol, and T/C ratio). Moreover, the relationships between free and total 25-(OH)D with psychophysical stress markers did not demonstrate the superiority of free over total measurements. Furthermore, training loads in different training periods did not affect resting vitamin D metabolite concentrations in football players.

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Should I stay or should I go pro? Early NFL draft entry by NCAA FBS underclassmen. Journal of Sports Economics, Ahead of Print: 1-23, 2024.

Abstract - Some college players face the tradeoff between continuing their education and entering the draft and forfeiting college eligibility. Little empirical research focuses on this "stay or go pro" decision. We analyze early draft entry decisions made by college football underclassmen with remaining eligibility over the 2007/08-2018/19 seasons. Regression results support both the human capital explanation for early draft entry in those players acquiring more human capital in the previous season by playing in more games increases the likelihood of early entry and the option value perspective in that playing on a more successful team increases the likelihood of early entry.

Sleep patterns before and after competition: A real-world examination of elite athletes. Journal of Sports Sciences, Ahead of Print: 1-14, 2024.

Abstract - Good sleep before and after competitions is crucial to cognitive, physiological performance and recovery. Yet, elite athletes face a unique set of challenges when acquiring good sleep before and after competitions, and indeed commonly report sleep problems when it matters most. This study examined the sleep of elite athletes before and after competition compared to before and after free days. A total of 1808 unique nights of actigraphy (n = 1495) and sleep diary (n = 1335) data from elite Australian Football League and National Rugby League male athletes (N = 85, M-age = 24.4 ± 3.6) were collected and analyzed using multi-level mixed models. On nights before competitions, athletes advanced sleep timings ($p < .001$, $d = 0.63$) and increased total sleep time ($p < .001$, $d = 0.65$) compared to nights before free days. On nights after competitions, athletes delayed sleep timings ($p < .001$, $d = 1.64$), reduced total sleep time ($p < .001$, $d = 1.28$), and had significantly worse quality sleep ($p < .001$, $d = 0.71$) compared to nights after free days. Sleep was especially worse following night competitions. While elite athletes and organizations may be implementing effective sleep strategies to optimize sleep the night before competitions, strategies to improve sleep after competitions may be lacking. We discuss potential factors contributing to this asymmetry and propose areas for research moving forward.

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Sudden cardiac death in young athletes – JACC State of the Art Review. Journal of the American College of Cardiology, Vol. 83(2): 350-370, 2024.

Abstract - Athletes epitomize the healthiest segment of society. Despite this premise, sudden cardiac death may occur in apparently healthy athletes, attracting significant attention not only in the medical community but also in laypersons and media. The incidence of sudden cardiac death is variably reported, and epidemiological burden differs among cohorts. Athletes appear to be at risk of developing fatal arrhythmias when harboring a quiescent cardiac disorder. Primary cardiomyopathies, ion channelopathies, and coronary artery anomalies are prevalent causes in young individuals. Cardiac assessment of athletes can be challenging because these individuals exhibit a plethora of electrical, structural, and functional physiological changes that overlap with cardiac pathology. A diagnosis of cardiac disease in a young athlete is not necessarily an indication to terminate competition and sports participation. International guidelines, traditionally focused on disqualification of individuals with cardiac disease, have recently adopted a more liberal attitude, based on a careful assessment of the risk and on a shared-decision making approach.

HIGHLIGHTS

- The incidence of SCD in young athletes has varied widely across studies. Inherited cardiac conditions and congenital abnormalities, such as coronary artery anomalies, are the most common etiologies of SCD in young athletes
- Early identification of disease in asymptomatic individuals through pre participation screening and policies aimed at implementing cardiopulmonary resuscitation in public spaces are means to prevent these events.
- A diagnosis of cardiac disease should not necessarily disqualify young athletes from participation in sports. Expert assessment and shared decision making should guide exercise prescription and engagement even in high-intensity sports

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The anabolic response to protein ingestion during recovery from exercise has no upper limit in magnitude and duration in vivo in humans. Cell Reports Medicine, 49(12):101324, 2023.

Abstract - The belief that the anabolic response to feeding during postexercise recovery is transient and has an upper limit and that excess amino acids are being oxidized lacks scientific proof. Using a comprehensive quadruple isotope tracer feeding-infusion approach, we show that the ingestion of 100 g protein results in a greater and more prolonged (>12 h) anabolic response when compared to the ingestion of 25 g protein. We demonstrate a dose-response increase in dietary-protein-derived plasma amino acid availability and subsequent incorporation into muscle protein. Ingestion of a large bolus of protein further increases whole-body protein net balance, mixed-muscle, myofibrillar, muscle connective, and plasma protein synthesis rates. Protein ingestion has a negligible impact on whole-body protein breakdown rates or amino acid oxidation rates. These findings demonstrate that the magnitude and duration of the anabolic response to protein ingestion is not restricted and has previously been underestimated in vivo in humans.

The delayed presentation of Achilles tendon ruptures is associated with marked alterations in the gene expression of COL1A1, MMOs, TIMPs, and IL-6. The American Journal of Sports Medicine, 52(1):164-173, 2024.

Background: Both acute and chronic Achilles tendon ruptures are affected by alterations in the extracellular matrix during the healing process of the tendon. Yet, these alterations in gene expression patterns are not well characterized.

Purpose: To characterize temporal and spatial differences in gene expression patterns after an Achilles tendon rupture and to evaluate if cells from chronic Achilles tendon ruptures have the same ability to form new tendon tissue (tendon constructs) as healthy tendon cells.

Study design: Controlled laboratory study.

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Methods: A total of 35 patients with surgically treated Achilles tendon ruptures were included in the study and divided into 3 groups: acute (<4 weeks), short-term chronic (1-6 months), and long-term chronic (>6 months). Biopsy specimens were collected during surgical repair and were used to analyze the gene expression within the different groups and to compare mRNA levels in the proximal and distal tendon ends. A complementary in vitro experiment was performed to evaluate if cells from chronic Achilles tendon ruptures can form tendon constructs.

Results: The mRNA levels for COL1A1 and COL3A1 were significantly higher in the short-term chronic group compared with the acute group ($P < .05$). Both MMP-1 and MMP-13 had the highest mRNA levels in the acute group ($P < .01$) compared with the long-term chronic group, while MMP-2 had the highest mRNA level in the short-term chronic group. Significant differences between the proximal and distal tendon ends were only detected for the monocyte and macrophage marker CD163 ($P < .05$), which was more expressed proximally. Cells extracted from chronic Achilles tendon ruptures displayed a similar ability and effectiveness to form tendon constructs as healthy tendon cells.

Conclusion: A high collagenase gene activity after an Achilles tendon rupture indicated possible rapid matrix degradation in the acute phase. Chronic ruptures appeared to initiate the healing process even before treatment, indicated by the higher expression of collagen in the short-term chronic group. Cells from chronic Achilles tendon ruptures also displayed an ability to form new tendon tissue in vitro.

Clinical relevance: The study shows a rapid increase in collagenase gene expression, which could lead to matrix degradation that continues for months after an Achilles tendon rupture.

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The effects of creatine supplementation on cognitive performance – A randomized controlled study. BMC Medicine, 21:440, 2023.

Background: Creatine is an organic compound that facilitates the recycling of energy-providing adenosine triphosphate (ATP) in muscle and brain tissue. It is a safe, well-studied supplement for strength training. Previous studies have shown that supplementation increases brain creatine levels, which might increase cognitive performance. The results of studies that have tested cognitive performance differ greatly, possibly due to different populations, supplementation regimens, and cognitive tasks. This is the largest study on the effect of creatine supplementation on cognitive performance to date.

Methods: Our trial was preregistered, cross-over, double-blind, placebo-controlled, and randomised, with daily supplementation of 5 g for 6 weeks each. We tested participants on Raven's Advanced Progressive Matrices (RAPM) and on the Backward Digit Span (BDS). In addition, we included eight exploratory cognitive tests. About half of our 123 participants were vegetarians and half were omnivores.

Results: Bayesian evidence supported a small beneficial effect of creatine. The creatine effect bordered significance for BDS ($p = 0.064$, $\eta 2P = 0.029$) but not RAPM ($p = 0.327$, $\eta 2P = 0.008$). There was no indication that creatine improved the performance of our exploratory cognitive tasks. Side effects were reported significantly more often for creatine than for placebo supplementation ($p = 0.002$, $RR = 4.25$). Vegetarians did not benefit more from creatine than omnivores.

Conclusions: Our study, in combination with the literature, implies that creatine might have a small beneficial effect. Larger studies are needed to confirm or rule out this effect. Given the safety and broad availability of creatine, this is well worth investigating; a small effect could have large benefits when scaled over time and over many people.

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The effects of repeated-sprint training on physical fitness and physiological adaptation in athletes: A systematic review and meta-analysis. Sports Medicine, Ahead of Print: 1-22, 2023.

Background: Repeated-sprint training (RST) is a common training method for enhancing physical fitness in athletes. To advance RST prescription, it is important to understand the effects of programming variables on physical fitness and physiological adaptation.

Objectives: This study (1) quantifies the pooled effects of running RST on changes in 10 and 20 m sprint time, maximal oxygen consumption (VO2max), Yo-Yo Intermittent Recovery Test Level 1 (YYIR1) distance, repeated-sprint ability (RSA), countermovement jump (CMJ) height and change of direction (COD) ability in athletes, and (2) examines the moderating effects of program duration, training frequency, weekly volume, sprint modality, repetition distance, number of repetitions per set and number of sets per session on changes in these outcome measures.

Methods: Pubmed, SPORTDiscus and Scopus databases were searched for original research articles up to 04 July 2023, investigating RST in healthy, able-bodied athletes, between 14 and 35 years of age, and a performance caliber of trained or above. RST interventions were limited to repeated, maximal running (land-based) sprints of ≤ 10 s duration, with ≤ 60 s recovery, performed for 2-12 weeks. A Downs and Black checklist was used to assess the methodological quality of the included studies. Eligible data were analyzed using multi-level mixed-effects meta-analysis, with standardized mean changes determined for all outcomes. Standardized effects [Hedges G (G)] were evaluated based on coverage of their confidence (compatibility) intervals (CI) using a strength and conditioning specific reference value of G = 0.25 to declare an improvement (i.e. G > 0.25) or impairment (i.e. G < - 0.25) in outcome measures. Applying the same analysis, the effects of programming variables were then evaluated against a reference RST program, consisting of three sets of 6 × 30 m straight-line sprints performed twice per week for 6 weeks (1200 m weekly volume).

Results: 40 publications were included in our investigation, with data from 48 RST groups (541 athletes) and 19 active control groups (213 athletes). Across all studies, the effects of RST were compatible with improvements in VO2max (G 0.56, 90% CI 0.32-0.80), YYIR1 distance (G 0.61, 90% CI 0.43-0.79), RSA decrement (G - 0.61, 90% CI - 0.85 to - 0.37), linear sprint times (10 m: G - 0.35, 90% CI - 0.48 to - 0.22; 20 m: G - 0.48, 90% CI - 0.69 to - 0.27), RSA average time (G - 0.34, 90% CI - 0.49 to - 0.18), CMJ height (G

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0.26, 90% CI 0.13-0.39) and COD ability (G - 0.32, 90% CI - 0.52 to - 0.12). Compared with the reference RST program, the effects of manipulating training frequency (+ 1 session per week), program duration (+ 1 extra training week), RST volume (+ 200 m per week), number of reps (+ 2 per set), number of sets per session (+ 1 set) or rep distance (+ 10 m per rep) were either non-substantial or comparable with an impairment in at least one outcome measure per programming variable.

Conclusions: Running-based RST improves speed, intermittent running performance, VO2max, RSA, COD ability and CMJ height in trained athletes. Performing three sets of 6 × 30 m sprints, twice per week for 6 weeks is effective for enhancing physical fitness and physiological adaptation. Additionally, since our findings do not provide conclusive support for the manipulation of RST variables, further work is needed to better understand how programming factors can be manipulated to augment training-induced adaptations.

Key Points

- Repeated-sprint training (RST) elicits moderate improvements in maximal oxygen consumption, Yo-Yo Intermittent Recovery Test Level 1 (YYIR1) distance and repeated-sprint ability (RSA) decrement, as well as small improvements in 10 and 20 m linear sprint times, RSA average time, countermovement jump height and change of direction ability.
- Compared with three sets per session, performing four sets per session may further enhance intermittent running performance (i.e. YYIR1 distance). Combined with a low number of repetitions (reps) (4–6 reps), this is a more effective training strategy to enhance physical performance rather than long series of exhaustive efforts (e.g. two sets of 10–12 reps).
- There was limited evidence to recommend increased training frequency (+1 session per week), program duration (+1 extra training week), RST volume (+200 m per week), number of reps (+2 per set) or rep distance (+10 m per rep) as beneficial to changes in physical qualities

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The impact of long-haul travel on the sleep of elite athletes. Neurobiology of Sleep and Circadian Rhythms, 15:100102, 2023.

Abstract - In order to manage and implement strategies to alleviate the symptoms of jet lag it is essential to assess the impact of jet lag in athletes. The aim of this study was to assess the impact of long-haul eastward travel on elite athletes' (n = 7 elite national track cyclists; male n = 3, and female n = 4) sleep. The athletes' sleep was monitored before, during and after travel using both actigraphy and self-report measures. Participants wore an activity monitor for 5 days prior to travel, during the long-haul travel and 5 days upon arrival at their destination and completed a daily online sleep diary Actigraphy highlighted significant reductions in time in bed, total sleep time and sleep efficiency (%) due to long haul eastward travel, particularly in the 48 h after travel. Sleep diary data exhibited significant reductions in time in bed, total sleep time, sleep efficiency, sleep quality and a significant increase in fatigue going to bed as a result of long-haul eastward travel. In order to facilitate the development of interventions to reduce the symptoms and severity of jet lag objective and subjective assessments of sleep should be coupled with assessments of chronotype and perceived sleep need.

The potential role of nutrition in overtraining syndrome: A narrative review. Nutrients, 15:4916, 2023.

Abstract - Competition between athletes and an increase in sporting knowledge have greatly influenced training methods while increasing the number of them more and more. As a result, the number of athletes who have increased the number and intensity of their workouts while decreasing recovery times is rising. Positive overtraining could be considered a natural and fundamental process when the result is adaptation and improved performance; however, in the absence of adequate recovery, negative overtraining could occur, causing fatigue, maladaptation, and inertia. One of the earliest forms of fatigue is overreaching. It is considered to be an accumulation of training that leads to reduced sports performance, requiring days or weeks to recover. Overreaching, if followed by adequate recovery, can lead to an increase in athletic performance. Nonetheless, if overreaching becomes extreme, combined with additional stressors, it could lead to overtraining syndrome (OTS). OTS, caused by systemic inflammation, leads to central nervous system (CNS) effects, including depressed mood, further

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inflammation, central fatigue, and ultimately neurohormonal changes. There are therefore not only physiological, biochemical, and immunological but also psychological symptoms or markers that must be considered, independently or together, being intrinsically linked with overtraining, to fully understand OTS. However, to date, there are very few published studies that have analyzed how nutrition in its specific food aspects, if compromised during OTS, can be both etiology and consequence of the syndrome. To date, OTS has not yet been fully studied, and the topic needs further research. The purpose of this narrative review is therefore to study how a correct diet and nutrition can influence OTS in all its aspects, from prevention to treatment.

The Tendinopathy Game Changers: Five papers from the last 4 years that might change how you manage tendons. Journal of Orthopaedic and Sports Physical Therapy, Vol. 54(1): 3-7, 2024.

SYNOPSIS - The clinical and scientific understanding of tendinopathy has substantially advanced since the Fifth International Scientific Tendinopathy Symposium in 2019. This editorial aims to highlight some of the fantastic tendinopathy research from the past 5 years. We have selected what we consider the "best paper" for each year from 2019 to 2023, which might change how you treat tendons. Selecting only 5 papers was not easy. Did your favorite papers make the cut? Or do you think we missed some key studies? We encourage you to tell us what you think using the social media hashtag #JOSPTtendon.

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Throwing cold water on muscle growth: A systematic review with meta-analysis of the effects of postexercise cold water immersion on resistance training-induced hypertrophy. European Journal of Sport Science, Ahead of Print: 1-13, 2024.

Abstract - The purpose of this paper was to systematically review the literature and perform a meta-analysis of the existing data on the effects of postexercise cold water immersion (CWI) coupled with resistance training (RT) on gains in measures of muscle growth. To locate relevant studies, we comprehensively searched the PubMed/MEDLINE, Scopus, and Web of Science databases. A total of 8 studies met the inclusion criteria; all investigated CWI as the means of cold application. Preliminary analyses conducted on noncontrolled effect sizes provided strong evidence of hypertrophic adaptations with RT that were likely to be at least small in magnitude (SMD0.5 = 0.36 [95% CrI: 0.10–0.61]; p (>0) = 0.995, p (>0.1) = 0.977). In contrast, noncontrolled effect sizes provided some evidence of hypertrophic adaptations with CWI þ RT that were likely to be small to negligible in magnitude (SMD0.5 = 0.14 [95% CrI: −0.08–0.36]; p (>0) = 0.906, p (>0.1) = 0.68). The primary analysis con ducted on comparative effect sizes provided some evidence of greater relative hypertrophic adaptations with RT compared to CWI þ RT (cSMD0.5 = −0.22 [95% CrI: −0.47 to 0.04]) with differences likely to be greater than zero (p (<0) = 0.957) and of at least a small magnitude of effect (p (<−0.1) = 0.834). Meta-regression did not indicate a potential moderation effect of training status ($_{\beta Trained:Untrained}$0:5 = −0.10 [95% CrI: −0.65 to 0.43] p < 0) = 0.653). In conclusion, based on the current data, the application of CWI immediately following bouts of RT may attenuate hypertrophic changes. Given the overall relatively fair to poor quality of the studies examined, the results of the current study should be interpreted with some caution.

Highlights

- Application of CWI immediately following bouts of RT may attenuate muscle hypertrophy in both trained and untrained individuals.
- The quality of evidence comparing RT versus CWI þ RT is generally fair to poor, indicating a need for higher quality studies on the topic to draw stronger inferences.

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Timing of anterior cruciate ligament surgery. Clinics in Sports Medicine, Ahead of Print: 1-11, 2023.

KEY POINTS

- There is considerable uncertainty surrounding the definition of early and delayed anterior cruciate ligament reconstruction (ACLR), with the definition for early surgery ranging from days to months and for delayed surgery ranging from weeks to years in the current literature.
- Modern arthroscopic surgical techniques and early rehabilitation have mitigated the risk of arthrofibrosis and postoperative stiffness with early ACLR.
- Early ACLR is shown to be equivalent or superior to delayed ACLR in terms of clinical outcomes, subjective postoperative knee function, the incidence of intra-articular injury within the timeframe between injury and surgery, as well as cost-effectiveness.
- Concomitant surgical procedures addressing anterolateral instability, anteromedial instability, and meniscus injury may be more advantageous when performed in conjunction with early ACLR.

Where does blood flow restriction fit in the toolbox of athletic development? A narrative review of the proposed mechanisms and potential applications. Sports Medicine, 53(11): 2077-2093, 2023.

Abstract - Blood flow-restricted exercise is currently used as a low-intensity time-efficient approach to reap many of the benefits of typical high-intensity training. Evidence continues to lend support to the notion that even highly trained individuals, such as athletes, still benefit from this mode of training. Both resistance and endurance exercise may be combined with blood flow restriction to provide a spectrum of adaptations in skeletal muscle, spanning from myofibrillar to mitochondrial adjustments. Such diverse adaptations would benefit both muscular strength and endurance qualities concurrently, which are demanded in athletic performance, most notably in team sports. Moreover, recent work indicates that when traditional high-load resistance training is supplemented with low-load, blood flow-restricted exercise, either in the same session or as a separate training block in a periodized program, a synergistic

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and complementary effect on training adaptations may occur. Transient reductions in mechanical loading of tissues afforded by low-load, blood flow-restricted exercise may also serve a purpose during de-loading, tapering or rehabilitation of musculoskeletal injury. This narrative review aims to expand on the current scientific and practical understanding of how blood flow restriction methods may be applied by coaches and practitioners to enhance current athletic development models.

Key Points

- Both resistance and endurance exercise may be combined with blood flow restriction to provide a broad range of adaptations in skeletal muscle, benefitting both muscular strength and endurance qualities concurrently, which are necessary for athletic performance in most sports.
- High-frequency blocks of blood flow restriction training may also selectively target satellite cells and promote myonuclear accretion, thereby priming skeletal muscle to attain greater hypertrophy in subsequent blocks of traditional high-load training.
- Practitioners and coaches may also use blood flow restriction exercise in their periodization models to supplement or de-load from traditional high-intensity training, or to accelerate recovery from musculoskeletal injury.
- Further research is warranted to optimize these approaches in athletic cohorts and to evaluate their practical merit.

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